Assignment Report for Assignment-04

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| Course and Section | 215.01 |
| Assignment Name | Assignment-04 |
| Due Date and Time | 08/16/2026 @ 11:59pm |
| | |
| First Name and Last Name | Ahmad Mustafa |
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| First Name and Last Name of Teammate | [First Name] [Last Name] |
| SFSU Email Account of Teammate | [SFSU Email Account] |


**PART A**

**Question Description and Analysis**:

2D Array, Single Data type.

**Answer**:

**Understanding the Problem:**

The program is asking me to create a pair of 2D arrays storing the english alphabet and to print it. One of the arrays to be created using the shorthand notation, and the other one created using the multi step approach.

The 2d must have minimum 5 rows and minimum 2 columns. One key detail to notice here is that the array must be irregular, which means that certain rows can have more than 2 columns, but not go lower than 2.

Another key detail from the sample output is that the alphabet is printed out in a right side alignment. This is important to know when setting up the print statements.

Talking about the print statements, the program requires both the shorthand notation and the multi step approach array to be printed out using one print method. This can easily be achieved by using a nested for-loop to go through the 2d array from left to right, and adding additional space as a placeholder.

**Post Programming reflection:**

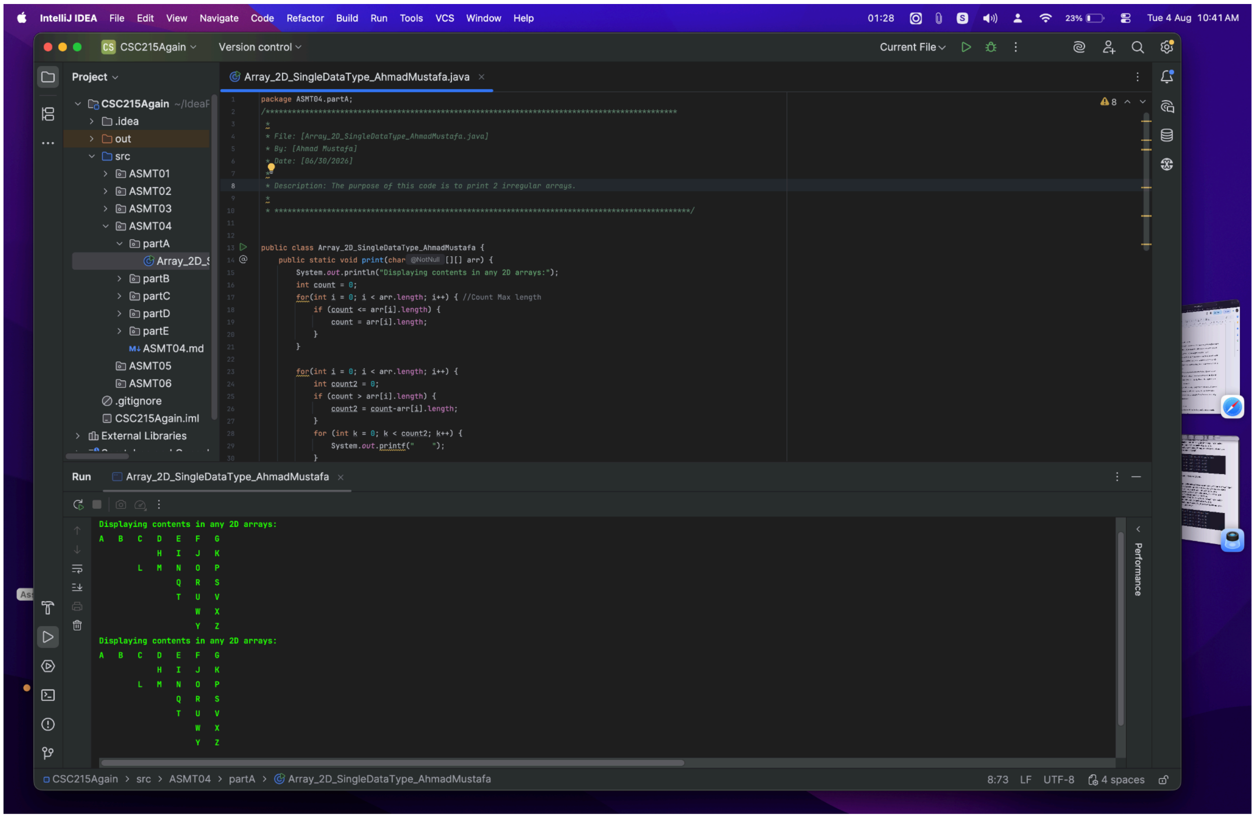
The coded solution Array_2D_SingleDataType_AhmadMustafa.java successfully compiles and creates the desired output.

One challenge while writing the code was to make it right aligned. I did not have much proficiency with printf, so the solution I came up with at the time was to fill the empty array addresses with a placeholder space. This helped me align the code to the right easily.

While writing this post programming reflection, I realized there is a grave error in the code; the arrays were made regular because of the placeholder spaces, while the client requires an irregular array.

Noticing this issue, I went back in the code to create both arrays as 'irregular' and created a new print method. This opened the challenge of making the program right aligned once more. But this time, equipped with a better understanding of the problem, I was able to right align by simply counting the row with the maximum columns, and then filling the rethe irregular array successfully as shown in the screenshots below.

**Screenshots of Outputs and Explanation**:



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**PART B**

**Question Description and Analysis**:

2D Array, Multiple Data Types

**Answer**:

**Understanding the Problem:**

Based off the first looks, the program is asking the user for 4 different inputs. First, it asks the user to enter:

- 3 integers,
- then 3 characters,

- then 3 strings
- and lastly, to enter 1 int, 1 char, and 1 string.

After receiving the inputs, the program prints the inputted data in 2 tables; First table prints out the data type names (int, char, string etc.) which must automatically be detected from the values. Second table prints out the actual values in the table.

In terms of implementation, the main requirement from the client is that the data must be stored in a 2D Array of Object data type. Overall, this is a basic take inputs, store inputs and print outputs program. The main challenge being faced here is to be able to efficiently store and print the data.

**Post Programming reflection:**

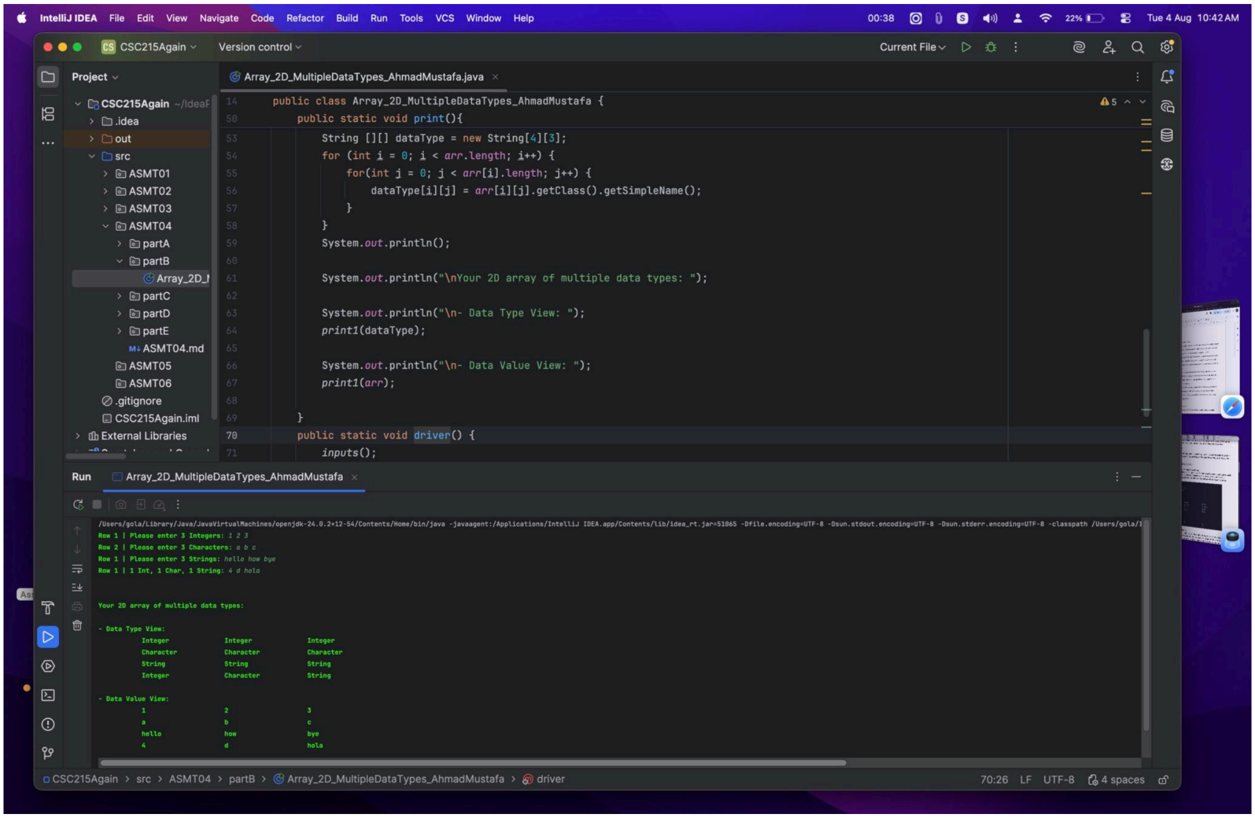
The program compiles successfully and produces the expected output. Compared to the sample output, the program is a 100% success.

Few challenges I faced while writing the program was my lack of knowledge on how I can print the data type name without literally typing it out. From a google search I discovered the *.getClass.getName();* method to extract the data type name from a variable or an array.

Using this I created a new array called dataType and individually scanned all the items in the main data array(arr) using a nested for loop. The result was an array called dataType with 4 rows and 3 columns. Each slot of the array the data type name. This new 2D array and the main data 2D array was printed using the print1() method I created to print any 2D array.

For the future, I would try to print the data types without storing it in a new array.

**Screenshots of Outputs and Explanation**:

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## PART C

**Question Description and Analysis**:

Gardening Planner

**Answer**:

**Understanding the Problem:**

The client is asking me to create a Plant growth calculator.

Firstly, I must ask the user; what are the ideal conditions for your plant to grow? Then based off of real world conditions which the client has provided, I must calculate how much the plant grows/regresses for each month of the year. Finally I must print out all the plant changes in the form of a table. This table has 6 columns:

1. Index
2. Month; (what is the name of the month)
3. Temperature; (what is the average temperature for this month?)
4. Rainfall; (what is the average rainfall for this month)
5. Plant Growth; (how much did the plant grow/regress in this month)
6. Plant Height; (what was the final height in this month)

The challenge in the program is to accurately calculate **5.** and **6.**

**Post Programming Reflection:**

Based off of the 3 test outputs the client provided, my program is a 100% success.

Firstly, I asked the user for the ideal conditions in order for the plant to grow. Using this data, I calculated the growth rate of the plant by comparing it to the actual conditions provided by the client.

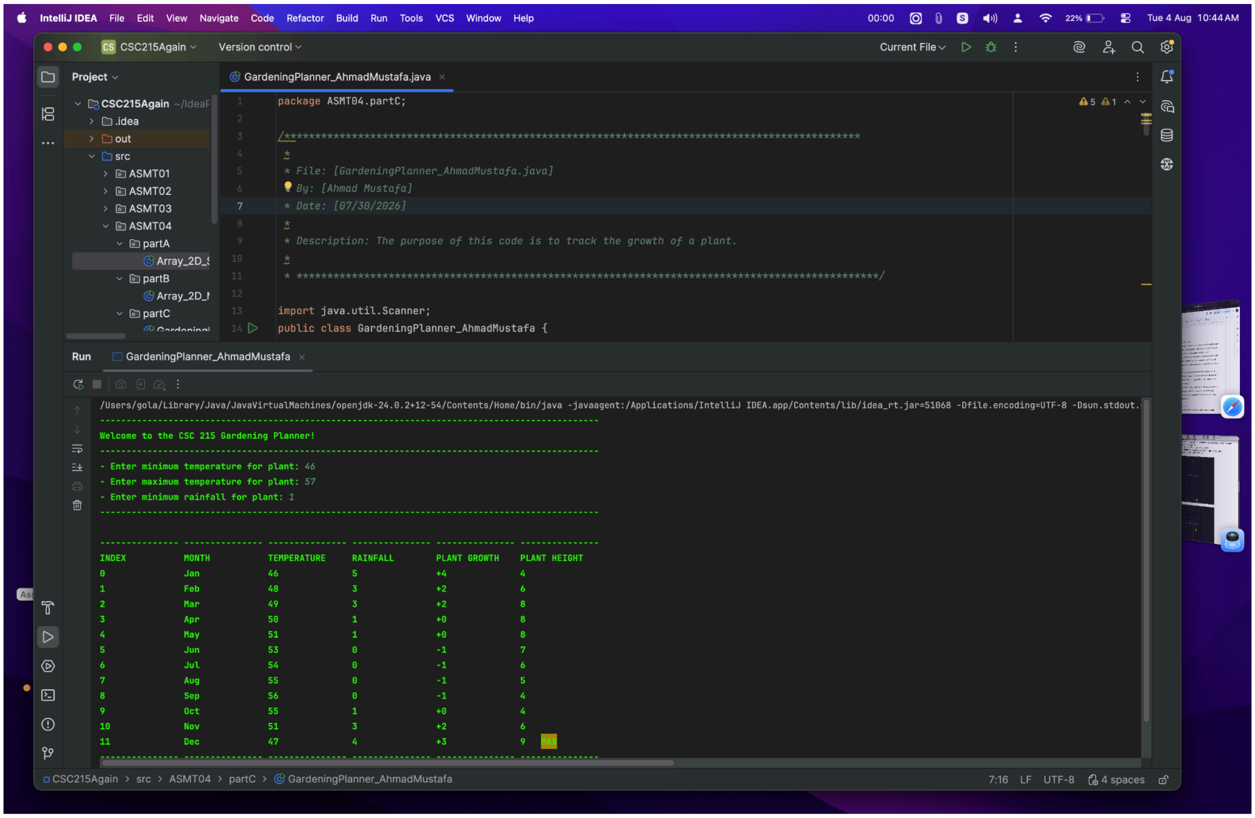
There are 2 calculations being made.

1. How much does the plant height increase or decrease <u>in meters</u>?

2. What is the <u>final height in meters</u> this month for the plant?

After making these 2 calculations for each month of the year, they are stored in a 2D array. This 2D array contains all the values needed to print out the final data table. Finally, I print out this 2D array to show the plant's progress throughout the year.

**Screenshots of Outputs and Explanation**:



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## PART D

### Question Description and Analysis:

Introductory Object Oriented Programming (OOP)

### Answer:

### Problem Analysis

The client wants the program to ask for the Name and GPA of 3 students, and then print this information. They also want to give their user an option to update previous student data. And after updating, they want to print that data to show the successful update. The client has provided us with a sample output for us to analyse.

In the sample output, the program is asking the Name and GPA for 3 students. After taking these inputs, the program prints out each of these student's information. Here the program prompts the user to enter the student they want to update. Then the user enters the full name of the student they want to update, this selects the student. Then the program prompts to enter the new Name and GPA for the student. Finally, the program outputs the 3 student's data once more and it finishes.

The challenge with this question is to accurately trace the student they want to update with their full name, and then update the records successfully.

**Post Programming Reflection**

The program meets all client requirements and produces expected output successfully. This has been made possible by using 4 methods. I will discuss these methods followed by the future update I would like to make for this program.

Starting with the 4 methods:

1. main()

This method contains one statement, which is to run the driver() method.

2. driver()

This method gives structure to the program by going through each step in order. It contains the print statements and runs the correct methods right after them. This method also allows the user to update student data, I'll touch more upon this later.

3. inputs(int count)

This method takes inputs from the user and stores them in the database.

It contains a loop to prompt student name and gpa, the amount of times is defined in the parameters. The user entries are directly stored in a Student[] array.

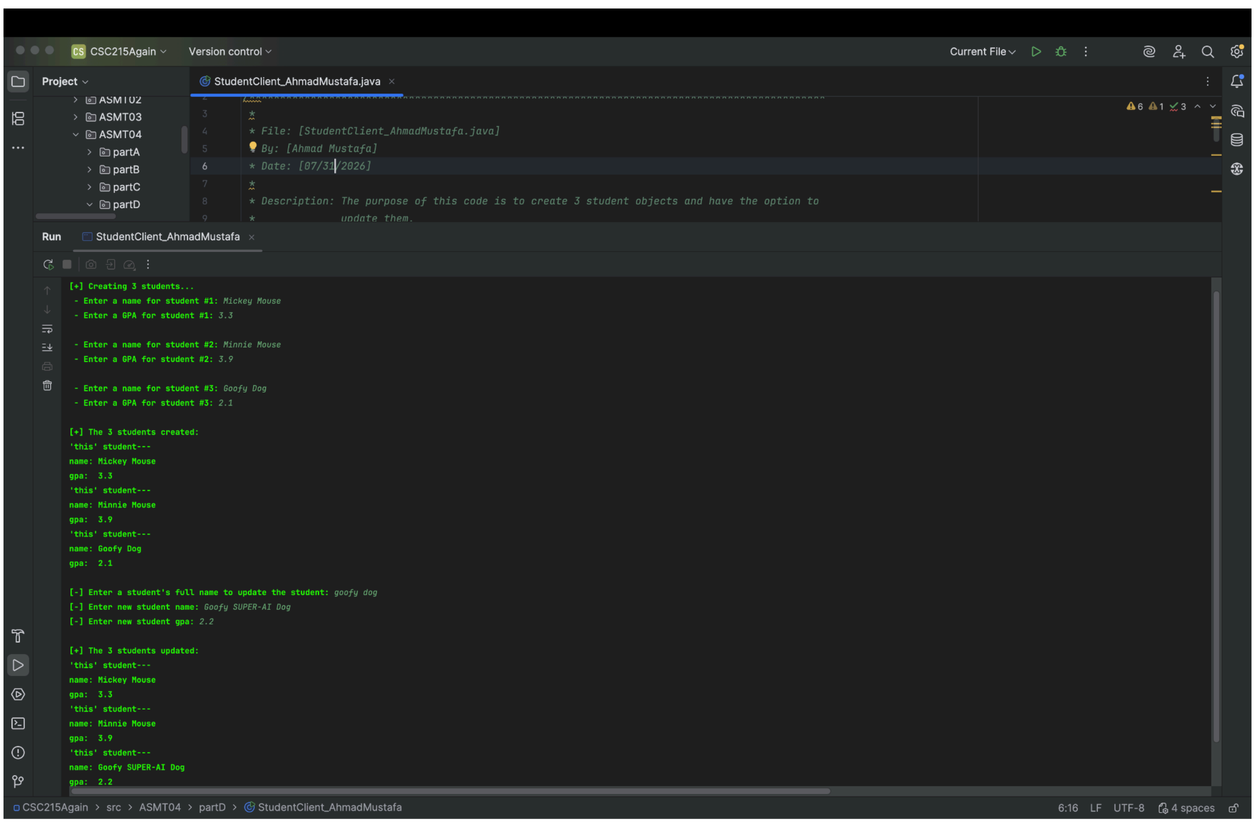
4. display()

This method displays the student data.

It contains a loop which print all Objects in the storage array.

Currently, there is no separate method for updating student data, which I would like to create in a future iteration of this program. This would allow for the students to be updated multiple times in a single instance. E.g. Do you want to update the students again? → Yes → studentUpdate()

**Screenshots of Outputs and Explanation**:



**PART E**

**Question Description and Analysis**:

3D Array, OOP Array

**Answer**:

Part A: How i loaded the 3D Array

The program is asking me to load 4 1D String arrays into a 3D String[4][3][2] array. The 4 1D String array contains 6 classes from a semester. Each 1D array represents a semester.

The way I'm planning to load this in the 3D array is as follows:

Each entry contains a class name. Each 1D array contains 2 class names. Each 2D array contains 3 1D arrays, hence 6 class names. And the 3D array contains 4 2D arrays, representing each semester.

To reiterate…

There are 4 semesters. Each semester has 3 Pairs of classes. Each pair is 2 classes. 4x3x2.

Part B: How I printed the 3D Array

I used a 3-nested loop to print this array. The challenge lies in entering a comma after every class except the last class of the semester. e.g. (class1, class2, class3,class4,class5, class6  class 1,...) I solved this challenge by using a modulus operator (%), since there are 6 classes in each semester, the program would NOT print a comma every 6 print outs. This ensures all classes have a comma right after them except the 6th, 12th, 18th and 24th. Making it seem as the text ends after each semester.

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Part A: How I loaded the objects

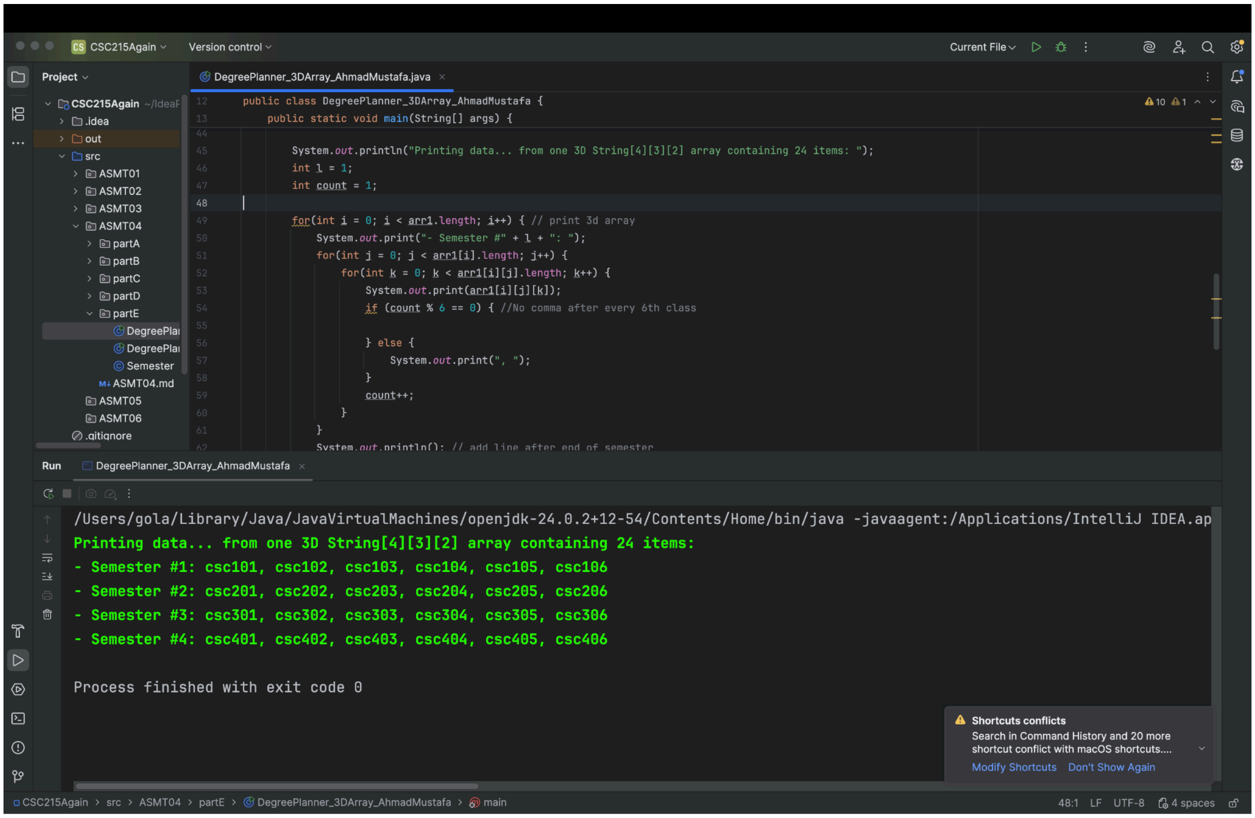
First I created a java class called Semester. There is one attribute within this class which is an array of Strings called "classes." This array contains 6 strings, each required to construct the object.

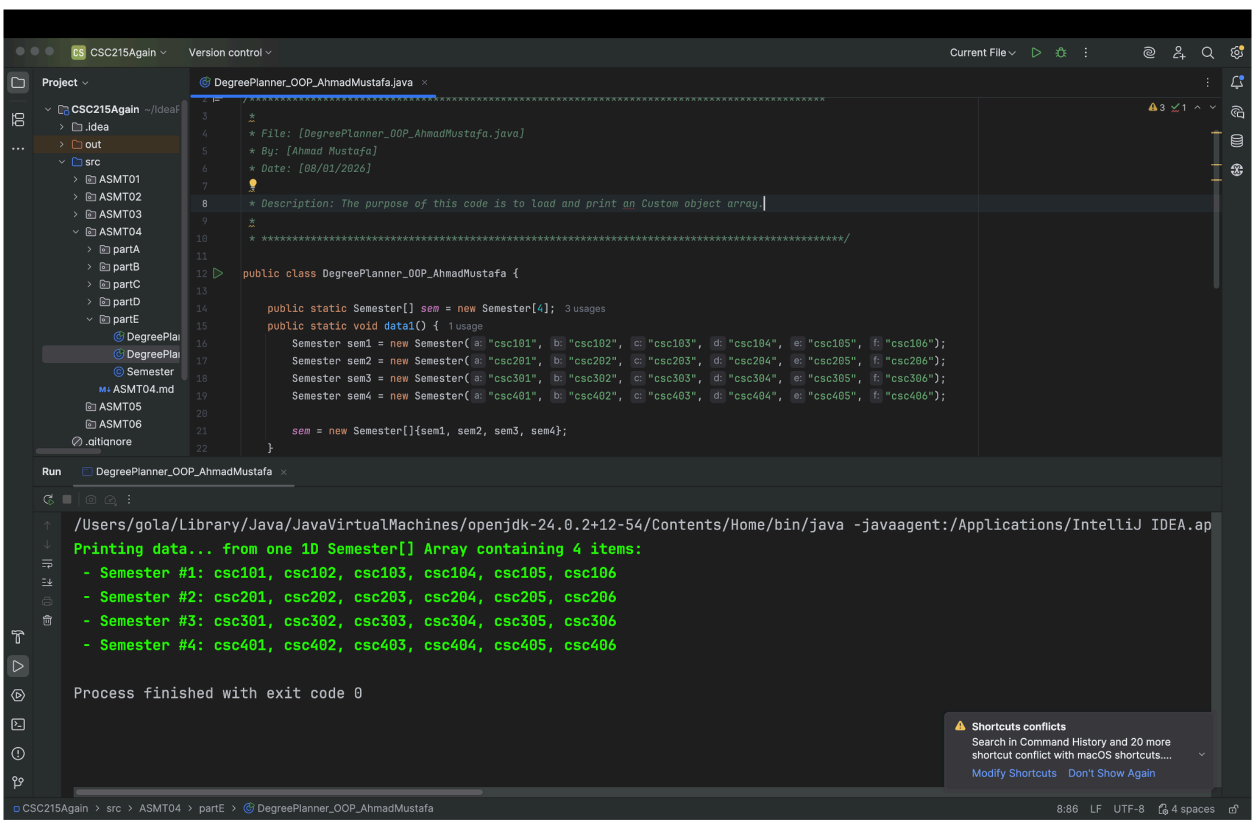
Secondly, I created a driver class to initialise 4 objects. These 4 Objects are of the Semester class created earlier, which are then stored in a 1D Semester array.

Part B: How I printed the objects

The print method of this version is similar to the 3d array version. The 6 classes within the semester are part of the class. I can simply use the object print method to print out the 6 classes within that object. Then in the driver class, I can loop 4 times to print each semester.

**Screenshots of Outputs and Explanation**:


CSC215Again
Version control
Current File
Project
DegreePlanner_3DArray_AhmadMustafa.java
CSC215Again ~/IdeaP
.idea
out
src
ASMT01
ASMT02
ASMT03
ASMT04
partA
partB
partC
partD
partE
DegreePla
DegreePla
Semester
ASMT04.md
ASMT05
ASMT06
.gitignore
public class DegreePlanner_3DArray_AhmadMustafa {
public static void main(String[] args) {
System.out.println("Printing data... from one 3D String[4][3][2] array containing 24 items: ");
int l = 1;
int count = 1;
for(int i = 0; i < arr1.length; i++) { // print 3d array
System.out.print("- Semester #" + l + ": ");
for(int j = 0; j < arr1[i].length; j++) {
for(int k = 0; k < arr1[i][j].length; k++) {
System.out.print(arr1[i][j][k]);
if (count % 6 == 0) { //No comma after every 6th class
} else {
System.out.print(", ");
}
count++;
}
}
System.out.println(); // add line after end of semester
Run
DegreePlanner_3DArray_AhmadMustafa
/Users/gola/Library/Java/JavaVirtualMachines/openjdk-24.0.2+12-54/Contents/Home/bin/java -javaagent:/Applications/IntelliJ IDEA.ap
Printing data... from one 3D String[4][3][2] array containing 24 items:
- Semester #1: csc101, csc102, csc103, csc104, csc105, csc106
- Semester #2: csc201, csc202, csc203, csc204, csc205, csc206
- Semester #3: csc301, csc302, csc303, csc304, csc305, csc306
- Semester #4: csc401, csc402, csc403, csc404, csc405, csc406
Process finished with exit code 0
Shortcuts conflicts
Search in Command History and 20 more shortcut conflict with macOS shortcuts....
Modify Shortcuts
Don't Show Again
CSC215Again › src › ASMT04 › partE › DegreePlanner_3DArray_AhmadMustafa › main
48:1 LF UTF-8 4 spaces


CSC215Again
Version control
Current File
Project
DegreePlanner_OOP_AhmadMustafa.java
*
* File: [DegreePlanner_OOP_AhmadMustafa.java]
* By: [Ahmad Mustafa]
* Date: [08/01/2026]
*
* Description: The purpose of this code is to load and print an Custom object array.
*
public class DegreePlanner_OOP_AhmadMustafa {
public static Semester[] sem = new Semester[4]; 3 usages
public static void data1() { 1 usage
Semester sem1 = new Semester( a: "csc101", b: "csc102", c: "csc103", d: "csc104", e: "csc105", f: "csc106");
Semester sem2 = new Semester( a: "csc201", b: "csc202", c: "csc203", d: "csc204", e: "csc205", f: "csc206");
Semester sem3 = new Semester( a: "csc301", b: "csc302", c: "csc303", d: "csc304", e: "csc305", f: "csc306");
Semester sem4 = new Semester( a: "csc401", b: "csc402", c: "csc403", d: "csc404", e: "csc405", f: "csc406");
sem = new Semester[]{sem1, sem2, sem3, sem4};
}
Run
DegreePlanner_OOP_AhmadMustafa
/Users/gola/Library/Java/JavaVirtualMachines/openjdk-24.0.2+12-54/Contents/Home/bin/java -javaagent:/Applications/IntelliJ IDEA.ap
Printing data... from one 1D Semester[] Array containing 4 items:
- Semester #1: csc101, csc102, csc103, csc104, csc105, csc106
- Semester #2: csc201, csc202, csc203, csc204, csc205, csc206
- Semester #3: csc301, csc302, csc303, csc304, csc305, csc306
- Semester #4: csc401, csc402, csc403, csc404, csc405, csc406
Process finished with exit code 0
Shortcuts conflicts
Search in Command History and 20 more shortcut conflict with macOS shortcuts....
Modify Shortcuts
Don't Show Again
CSC215Again › src › ASMT04 › partE › DegreePlanner_OOP_AhmadMustafa
8:86 LF UTF-8 4 spaces